

MATTER IN OUR SURROUNDINGS

• Introduction

“Chemistry”

Derived from

‘Chemia’ [Ancient word of Mishra]

- Chemica means “Black Colour”.
- The famous name of Chemia was “Chemi”.

Note:

1. The Black Smooth Soil is called “Chemia” by Egyptians.
2. The Greeks also called Chemia after Egyptians.
3. The ancient name of Chemistry was “Chemiteching”.

- According to Roald Hoffmann, Chemistry is the science of “atoms and molecules” and their transformation. It is science not so much of one hundred elements but of the infinite variety of molecules that may be build from them.
- Chemistry is also called science of “atoms and molecules”.
- Chemistry is chiefly concerned atoms and molecules and their inter – action and transformation.
- Chemistry is also plays an important role in our daily life.

• Definition

- Chemistry is the science concerned with the composition, structure, and properties of matter as well as the changes it undergoes during chemical reaction.

• Modern Definition

- Chemistry is an experimental science in which we study about atoms and molecules and their transformation.
- The father of Chemistry – Antonie D. Lavoisier.

• Matter

- Anything/Everything which occupies space, has mass and creates resistance is called matter.
OR
- Everything we can see and touch is called matter.

For Example:-

- ✓ Pen, Book, Chair, Human, Cooler, House etc.
- Matter is made up of very small particle called atoms, hence matter is not continuous but particulate in the nature.

• Continuous

- Does not have any brakes i.e, present in single form.
- Every matter is made up of certain particles which differ in shape, size and nature from other types of matter.

Note:

- Light, Heat, Sound, Electricity, energy are not matter because they do not have any mass and they do not occupied space.

- **Nature/Physical properties of matter**

- Matter is made up discrete particles.
- The particles of matters attracts to each other.
- There are empty space between particles of matter.
- The particles of matter are too small to be imagined.
- The particles of some matter are in constant motion.

- **Substance**

- A substance is the purest form of matter which can't be broken into two or more parts by any physical process.

For Example:-

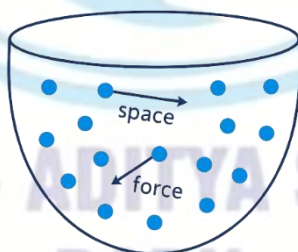
- ✓ Sugar, Salt, Lime etc.

- **Inter – molecular force of attraction**

- The Force due to which particles of matter are held together, called inter – molecular force of attraction.
OR
- The Force of attraction between particles of matter due to which they bonded together, called inter – molecular force of attraction.

- **Inter – Molecular Space**

- The empty space between particles of matter is called inter – molecular space.



Inter – molecular Space/ Force

- **Classification of Matter**

1. On the basis of physical properties.
2. On the basis of Chemical properties.

└───────────> In next Chapter

- **Case 1:**

- On the basis of physical properties.

- (i) Solid State of matter.
- (ii) Liquid State of Matter.
- (iii) Gaseous State of Matter.

- The fourth State of matter → “Plasma State”.

➤ The fifth State of matter → The Bose – Einstein condensate [B.E.C]

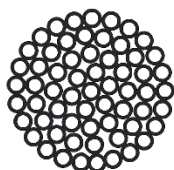
• **Solid State of matter**

For Example:-

- ✓ Stone, bricks, pen, chair, table, board, iron, gold etc.

Definition:-

- That state of matter in which, molecules/atoms particle are closely packed and are being held together by very strong forces.
- The inter molecular force of attraction is very high due to this reason solids are very strong.
- Inter – molecule space is very very minimum due to this reason particles of solid can't move (immovable) particles of matter can only vibrates on their original position.



The particles of solid matter

• **Properties of Solid Substances**

- (i) Solid substances have fixed shape and volume.

For Example:-

- ✓ Book, table, chair, pen etc.

- (ii) The density of solid substances is high.

$$\text{i.e, density} = \frac{\text{mass}}{\text{volume}} \quad \left\{ \begin{array}{l} \text{Density} \propto \text{mass} \\ \text{Density} \propto \frac{1}{\text{volume}} \end{array} \right.$$

- (iii) The melting and boiling point of solids are high.

- (iv) Solids are hard and rigid.

- (v) Solids are incompressible i.e, it can't be compressed.

- (vi) Solids having no flowing properties.

- (vii) Solids have no diffusion properties. Exception – black board & Chalk particle diffusion, iron articles diffusion after long time.

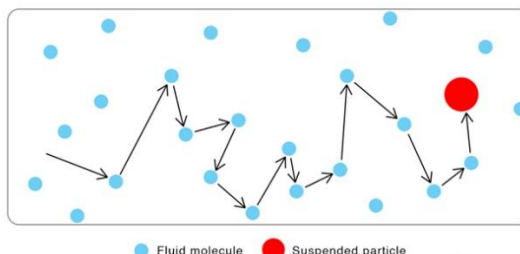
- (viii) When Solids are either heated or cooled. There is no more effect on expansion or contraction of solids.

Note:

1. Solids have very low diffusion properties.
2. Solids are very slightly compressible in mostly incompressible.

• **Brownian Motion**

- Brownian motion is the random, zig-zag movement of microscopic particles (like dust or pollen) suspended in a fluid (liquid or gas).



● Fluid molecule ● Suspended particle

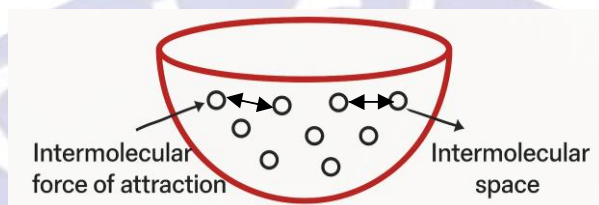
- **Liquid State of Matter**

For Example:-

- ✓ Water, Milk, Oil etc.

Definition:-

- That state of matter in which, particles (atom/molecules) are less tightly packed as compared to solids.
- The inter – molecular force of attraction is less than the solids due to this reasons particles are bounded by weak bond.
- The inter – molecular space is more than solid due to this reason particles move freely from one place to another place.



- **Properties of Liquid Substances**

- Liquid have no fixed shape but have fixed volume.
- The density of liquids is less than solids.
- The melting and boiling point of liquids is less than solids.
- Liquids are incompressible.
- Liquids have flowing tendency.
- Liquids have diffusion tendency.
- When Liquids are heated, then they forms vapour and this process is known as vaporation.
- When Liquids like as water is cooled to 0°C (273K) then it forms ice and this process is known as freezing of liquid.

Note:

- Liquids are very very slightly compressible.

- **Gaseous State of Matter**

- ✓ L.P.G, C.N.G, Natural Gas, Air etc.

Definition:-

- That state of matter in which particles (atoms/molecules) are very less tightly packed as compared to solids as well as liquids.
- The intermolecular force of attraction between particles of gas less than liquid as well as solid.
- The intermolecular space between the particles is more than the solids as well as liquids due to this reason particles of gas are bonded by very weak bond.
- The particles of gases can easily move from one space to another space, whatever particles comes out from the gas.



Highly
moveable
particles
(max
intermolecular
space)
(min
intermolecular
force)

- **Properties of gaseous substances**

- Gases have neither fixed shape nor fixed volume.
- The density of gases is less than solids as well as liquids.
- The melting and boiling points of gases is less than normal temperature or room temperature (25°C).
- Gases are very compressible.
- It is expand or contract by heating or cooling.
- It has diffusion tendency.
The gaseous form of matter changes into liquid form by using compression.
- Pressure is also created by gas.

- **Diffusion**

- That process in which the particles of a substance is diffused in the particles of other substance.

- **Case 1:** Solid – Solid Diffusion

- The particles of solids do not diffused in the particles of other solids. But, in some cases the particles of solid diffused partially in the particles of other solid.

For Example:-

- ✓ When two iron rods are bind together with a metallic wire and left it for a long time. Then, particles of both rods partially diffused and make a bond in outer surface.
- ✓ When Chalk particles are left for some days on black board then chalk particles diffused with black board particles.

- **Case 2:** Solid – liquid diffusion

- The particles of solids are diffused in the particles of liquids.

For Example:-

- ✓ The particles of solids like as sugar, salt, iodine etc are diffuse in the particles of water.

- **Case 3:** Gas – Gas Diffusion

- The particles of gas are diffused in the particles of other gas.

For Example:-

- ✓ In Air many gases like as Nitrogen, Oxygen, Carbon dioxide are diffused.

- **Case 4:** Liquid – Liquid Diffusion

- The particles of liquid are diffused in the particles of other liquid.

For Example:-

- ✓ The particles of water are diffused in the particles of milk.
- ✓ Kerosene is mixed with diesel and petrol.
- ✓ The L.P.G is diffused in C.N.G.

- **Case 5:** Liquid – Gas diffusion

- The particles of gases are diffused in the particles of liquid.

For Example:-

- In water (liquid), Oxygen and Hydrogen both gases are diffused.

- **Sublimation**

- That process in which, the change of the solid state of matter directly on heating to vapour state without becoming liquid state, called sublimation.

For Example:-

- ✓ Ammonium Chloride → NH_4Cl
- ✓ Camphor → $\text{C}_{10}\text{H}_{16}\text{O}$
- ✓ Naphthalene → C_{10}H_8
- ✓ Iodine → I_2
- ✓ Dry Ice/Solid CO_2 → CO_2

Note:

- These also when cooled it directly change into solid without change into liquid.

• **Evaporation**

- That process in which a liquid changes into vapour at all temperature (either room temperature or below B.P), from the surface of the liquid, called evaporation.

▪ **Cause of affecting of evaporation**

Case 1: On the Nature of liquid

- evaporation is inversely proportional to boiling point of liquid

i.e, $\text{evaporation} \propto \frac{1}{\text{B.P of liquid}}$

For Example:-

- ✓ Ether (34°C) more evaporates than water at room temperature.

Case 2: On the temperature of the liquid

- Evaporation is directly proportional to the temperature.

i.e, $\text{evaporation} \propto \text{temperature}$

Case 3: On the surface area of the liquid's container

$\text{evaporation} \propto \text{Surface area of liquid's container}$

Case 4: On the velocity of the air

$\text{evaporation} \propto \text{Velocity of Air (Dry)}$

Case 5: On the humidity of air

$\text{evaporation} \propto \frac{1}{\text{humidity}}$

For Example:-

- ✓ For drying cloth takes more time in humid air than dry air.

• **Dry air**

- The amount of vapour particles less than a fixed value then it is said to be dry air.

- **Humid air**

- The amount of water vapors present in the air is known as Humid air.

For Example:-

- ✓ Fog Containing air.

- **Boiling point (B.P)**

- That minimum temperature at which any substance start to boil is known as boiling point of that liquid.

For Example:-

- ✓ B.P of water = 100°C
- ✓ B.P of ether = 34°C
- ✓ B.P of alcohol = 78.3°C (78°C)

- **Cause of effect of boiling**

Case 1: In the presence of impurities

- Boiling point of liquid is directly proportional to impurities present in it.
$$\text{B.P} \propto \text{Impurities Presence}$$

For Example:-

- ✓ In (Water + Salt) solutions, B.P (↑) due to salt forms ionic bonds.
- ✓ In (Water + Sugar) solutions, B.P (↑) due to Sugar forms covalent bond.

Case 2: On the pressure

- Since a liquid vaporises when molecules move faster and further apart but while increasing pressure, you can applying an external force to hold molecules closer to each – other.
$$\text{B.P} \propto \text{Pressure}$$

For Example:-

- ✓ The B.P of water in pressure cooker is more than 100°C [Approx 121°C]

Note:

1. Pressure (↑), B.P (↑)
Pressure (↓), B.P (↓)
2. Cooking food on mountain takes more time as compared to earth's surface due to low pressure on the mountain.

- **Melting point**

- That temperature at which any solids melts into liquid is known as melting point of that solid.

For Example:-

- ✓ Melting point of ice = 0°C
- ✓ Melting point of iron = 1538°C
- ✓ Melting point of Tungston = 3422°C
- ✓ Melting point of aluminium = 660.3°C

- **Affecting of pressure on melting**

- The melting point of most of the solid substance increases with an increase in pressure.
but

Case 1: Those substances which volume increases on melting, their melting point increases when increase in pressure.

For Example:-

- ✓ Wax, Copper etc.

Case 2: Those substances which volume decreases on melting, their melting point decreases when increase in pressure.

For Example:-

- ✓ Ice.

- **Freezing point**

- That constant temperature at which any liquid starts to freeze and becomes in solid is called freezing point of that liquid.

For example:-

- ✓ Freezing point of water is 0°C.

- **Factors affecting of freezing**

1. Freezing Point $\propto \frac{1}{\text{Impurities present in the liquid}}$
- ✓ Like as Salt, Sugar etc.
2. Freezing Point $\propto$ Inter molecular force of attraction between particles.
3. Freezing Point $\propto$ Pressure of substance
- ✓ Exception → In the case of water F.P decreases.

DIFFERENCE BETWEEN EVAPORATION & BOILING

Evaporation	Boiling
Evaporation is a spontaneous process which occur at all temperature.	Boiling occur at a fixed temperature.
It occur only outer surface of liquid.	It occurs all surface of the liquid.
It is a silent process.	It is a noisy process.
It do not form bubbles.	It forms bubbles.
It shows cooling effect.	It don't show cooling effect.

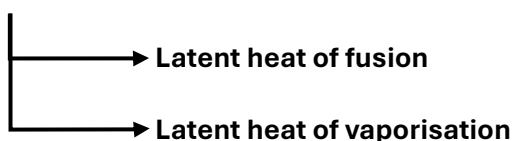
- **Latent Heat**

- The heat required to change the state of matter is known as latent heat.
- The S.I unit of latent heat is J/kg.

Note:

- The heat required only from atmosphere.

Latent heat



- **Latent heat fusion**

- The heat required to 1 kg of solid to completely change into liquid is known as latent heat of that solid fusion.

For Example:-

- ✓ The latent heat of ice = 80 cal/g.

$$= 3.345 \times 10^5 \text{ J/Kg}$$

i.e, 3.345×10^5 Joule heat required to completely change 1 kg ice into liquid.

▪ **Latent heat of vaporization**

- The heat required to change 1 kg of liquid completely change into vapour is known as latent heat of vaporization.

For Example:-

- ✓ The latent heat of water = 540 cal/g.

$$= 22.59 \times 10^5 \text{ J/kg}$$

i.e, 22.59×10^5 J energy required to change 1 kg of liquid into vapour.

DIFFERENCE BETWEEN VAPOUR AND GAS

Vapour	Gas
It is homogeneous mixture of liquid particles in suspension with true gas of the same substance.	It is the state of substance in which the evaporation from the liquid state is complete.
Vapour do not obey gas law.	Generally, gas obey the gas law.
Vapour occurred at any temperature.	It is state of matter.
The natural state of vapour is either liquid or gas.	Gas is its natural state at room temperature (25°C)

Q. The vapour of a liquid at 100°C is more harmful than boiled water at 100°C Why? Explain it.

Ans The vapour at 100°C (Steam) is associated with more energy due to the high specific latent heat of vaporization causes more serious burns than water at 100°C.

• **Freezing Mixture**

- When we adding some salt or sugar in manufacturing of ice. Then its freezing point less upto -22°C. This type of mixture is known as freezing mixture.

For Example:-

- ✓ Ice – cream, Softy, etc.

• **Critical temperature [T_c]**

- The critical temperature of a substance is defined as, “The highest temperature at which the substance can exist as a liquid at above the critical temperature the substance in its vapour/gaseous state can no longer be liquified, regardless of the amount of pressure applied to it.

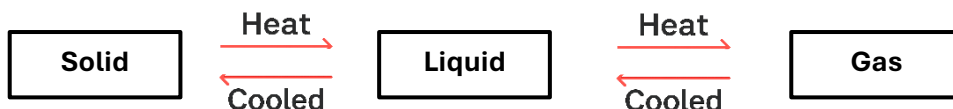
i.e, The critical temperature of a substance is the temperature at and above which vapour of the substance can't be liquified, no matter how much pressure is applied.

For Example:-

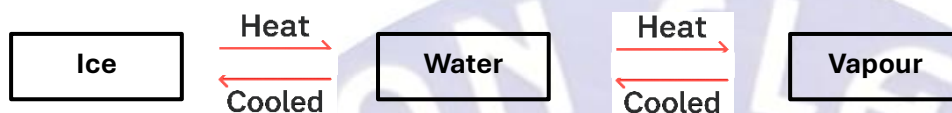
Substance	Critical pressure (atm)	Critical temperature
NH ₃	115.5	132°C
O ₂	49.7	-119°C
CO ₂	73.0	31.2°C
H ₂ O	217.7	374°C

Or, The temperature above which gas can't be liquified no matter how great the pressure.

• **Inter convertibility of state of matter**



For Example:-



▪ **The fourth state of matter**

- A new type of state of matter is discovered which is known as Plasma state of matter.
In Plasma state, the particles of matter is in super energetic and super excited form.
- Plasma is an electrically charged gas. Because plasma particles have an electric charged, they are affected by electric and magnetic fields. Thus, Plasma is differ from a gas.
- Plasma is a mixture of free electrons and ions.
- The sun and other stars glow because of the presence of Plasma in them.
- Plasma can also be made on the earth by passing electricity through gases at very low pressure taken in a glass tube (called discharged tube)

For Example:-

- ✓ Sun, Stars, neon filled bulb, CFL mercury etc have more brighter due to plasma.

Note:

- The fourth state of matter is also called "Plasma State".

▪ **The fifth state of matter**

- Before, 19 century Einstein predicates that the fifth state of matter exist in the nature.
In 1920 the Indian famous scientist Satendra Nath Bose defined the fifth state of matter, this state of matter is called Bose Einstein Condensate [B.E.C].
- For obtain this state the density decreases 10^5 times with respect to air and so much lighter density gas was super cooled at a temperature.
- Thus, the fifth state of matter is Bose Einstein Condensate is also called B.E.C.
- The B.E.C state is further studies in higher classes.

Q. Why do earthen pot contain cool water?

Ans There are many small pores in an earthen pot through these pores the water kept inside continuously keep seeping through these pores to outside the pots and these water droplets directly contact with atmosphere and evaporates. It takes the latent heat required for vaporization from the earthen pot inside water. Hence, water loses that heat making the water inside the pot cool.

Q. Why do we wear cotton clothes in the summer?

Ans We know that cotton clothes easily absorbed sweat from the body. Cotton clothes have fine threads which easily allow air to pass through when compared to other clothes. When more air flowing through the clothes, the sweat from the body and clothes gets evaporated easily, which leaves the cooling effect.